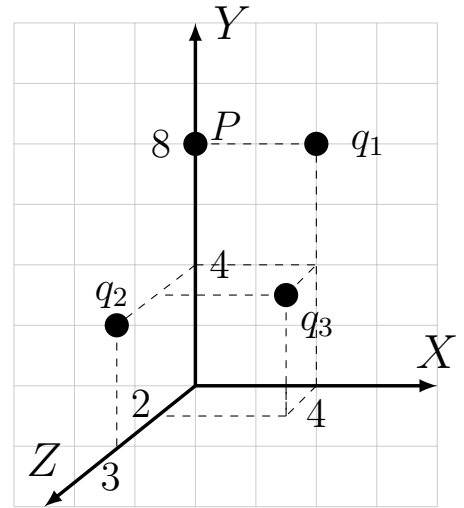


1. (19 points) Se dispone de un sistema formado por 3 cargas puntuales: $q_1 = 8[nC]$, $q_2 = -25[nC]$ y $q_3 = 24[nC]$ ubicadas en los puntos $P_1(4, 8, 0)$, $P_2(0, 4, 3)$ y $P_3(4, 4, 2)$ respectivamente. Considere que las posiciones de las cargas están en centímetros y que el valor de $K = 9 \times 10^9 [Nm^2/C^2]$, responda

Nota:

prefijos: $c = 10^{-2}$, $m = 10^{-3}$, $\mu = 10^{-6}$, $n = 10^{-9}$, $p = 10^{-12}$.



- (a) (8.0 pts) calcule el potencial eléctrico en un punto $P(0, 8, 0)$

$$\begin{aligned} V_P &= V_{1P} + V_{2P} + V_{3P} \\ &= \frac{9 \cdot 10^9 (8 \cdot 10^{-9})}{4 \cdot 10^{-2}} + \frac{9 \cdot 10^9 (-25 \cdot 10^{-9})}{5 \cdot 10^{-2}} + \frac{9 \cdot 10^9 (24 \cdot 10^{-9})}{6 \cdot 10^{-2}} \quad (1,0 \text{ pts}) \\ &= (18 - 45 + 36) \cdot 10^2 = 9 \cdot 10^2 = 900 [V] \quad (2,0 \text{ pts}) \end{aligned}$$

$$\vec{r}_{1P} = \vec{r}_P - \vec{r}_1 = 8\hat{y} - (4\hat{x} + 8\hat{y}) = -4\hat{x} \text{ cm} \Rightarrow \|\vec{r}_{1P}\| = 4 \cdot 10^{-2} [m] \quad (1,0 \text{ pts})$$

$$\vec{r}_{2P} = \vec{r}_P - \vec{r}_2 = 8\hat{y} - (4\hat{y} + 3\hat{z}) = (4\hat{y} + 3\hat{z}) \text{ cm} \Rightarrow \|\vec{r}_{2P}\| = 5 \cdot 10^{-2} [m] \quad (1,0 \text{ pts})$$

$$\vec{r}_{3P} = \vec{r}_P - \vec{r}_3 = 8\hat{y} - (4\hat{x} + 4\hat{y} + 2\hat{z}) = -4\hat{x} + 4\hat{y} - 2\hat{z} \text{ cm} \Rightarrow \|\vec{r}_{3P}\| = 6 \cdot 10^{-2} [m] \quad (1,0 \text{ pts})$$

- (b) (8.0 pts) calcule la energía contenida en el sistema.

$$U_1 = V_0 q_1 = 0 [J] \quad (1,0 \text{ pts})$$

$$U_2 = V_1 q_2 = \frac{k q_1 q_2}{\|\vec{r}_{12}\|} = \frac{9 \cdot 10^9 (8 \cdot 10^{-9}) (-25 \cdot 10^{-9})}{\sqrt{41} \cdot 10^{-2}} = -281,11 \cdot 10^{-7} [J] = -28,11 [\mu J] \quad (1,0 \text{ pts})$$

$$\begin{aligned} U_3 &= (V_1 + V_2) q_3 = \frac{k q_1 q_3}{\|\vec{r}_{13}\|} + \frac{k q_2 q_3}{\|\vec{r}_{23}\|} = \frac{(9 \cdot 10^9) (8 \cdot 10^{-9}) (24 \cdot 10^{-9})}{\sqrt{20} \cdot 10^{-2}} + \frac{(9 \cdot 10^9) (-25 \cdot 10^{-9}) (24 \cdot 10^{-9})}{\sqrt{17} \cdot 10^{-2}} \\ &= (-923,310^{-7} [J]) = -92,33 [\mu J] \quad (1,0 \text{ pts}) \end{aligned}$$

$$U = U_1 + U_2 + U_3 = (0 - 28,11 - 92,33) \cdot 10^{-6} = -1204,4 \cdot 10^{-7} = -120,44 [\mu J] \quad (2,0 \text{ pts})$$

$$\vec{r}_{12} = \vec{r}_2 - \vec{r}_1 = 4\hat{y} + 3\hat{z} - (4\hat{x} + 8\hat{y}) = (-4\hat{x} - 4\hat{y} + 3\hat{z}) \text{ cm} \Rightarrow \|\vec{r}_{12}\| = \sqrt{41} \text{ cm} = 6,4 \cdot 10^{-2} [m] \quad (1,0 \text{ pts})$$

$$\vec{r}_{13} = \vec{r}_3 - \vec{r}_1 = 4\hat{x} + 4\hat{y} + 2\hat{z} - (4\hat{x} + 8\hat{y}) = (-4\hat{y} + 2\hat{z}) \text{ cm} \Rightarrow \|\vec{r}_{13}\| = \sqrt{20} \text{ cm} = 4,47 \cdot 10^{-2} [m] \quad (1,0 \text{ pts})$$

$$\vec{r}_{23} = \vec{r}_3 - \vec{r}_2 = 4\hat{x} + 4\hat{y} + 2\hat{z} - (4\hat{y} + 3\hat{z}) = (4\hat{x} - \hat{z}) \text{ cm} \Rightarrow \|\vec{r}_{23}\| = \sqrt{17} \text{ cm} = 4,12 \cdot 10^{-2} [m] \quad (1,0 \text{ pts})$$

- (c) (3.0 pts) calcule el trabajo necesario en traer una carga $q_4 = -7[nC]$ al punto P.

$$\begin{aligned} W &= U = q_4 V_P \\ &= -7 \cdot 10^{-9} (900) \quad (1,0 \text{ pts}) \\ &= -63 \cdot 10^{-7} \\ &= -6,3 [\mu J] \quad (2,0 \text{ pts}) \end{aligned}$$